

# CSC320 Lab1 Report

hongcheng huo

October 2026

Organize the PDF by task. Screenshots must be large enough for us to see the selected points, lines, labels, and numerical values in the interactive tasks.

## Task 1 PDF contents

- Your three photographs: `mild.jpg`, `strong.jpg`, and `failure.jpg`
- Screenshots of the interactive affine and homography canvas showing:
  - The selected outer corners
  - The selected held-out inner corners
  - Both the mild and strong viewpoints
- The rectification baseline comparison plot
- The corner jitter diagnostic plot
- The corner uncertainty results plot
- `failure.jpg` with its four selected corners visible
- The rectified output for `failure.jpg`
- Your written answers for Tasks 1.2, 1.3, and 1.4

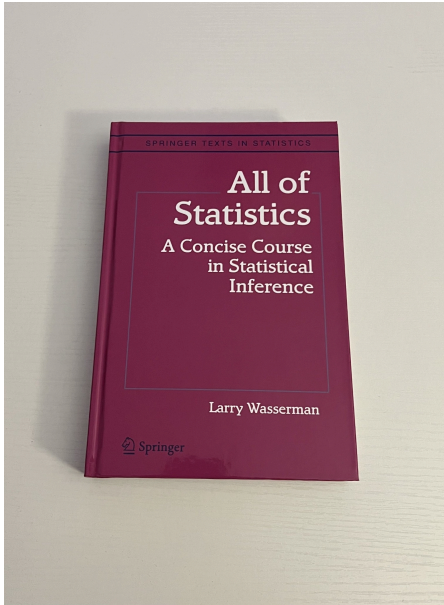
## Task 2 PDF contents

- A screenshot of the interactive canvas showing:
  - Both selected line groups
  - The fitted vanishing points
  - The estimated horizon
- Your derivations and answers for Tasks 2.2, 2.3, 2.4, and 2.8
- Your final focal-length results in pixels and millimetres

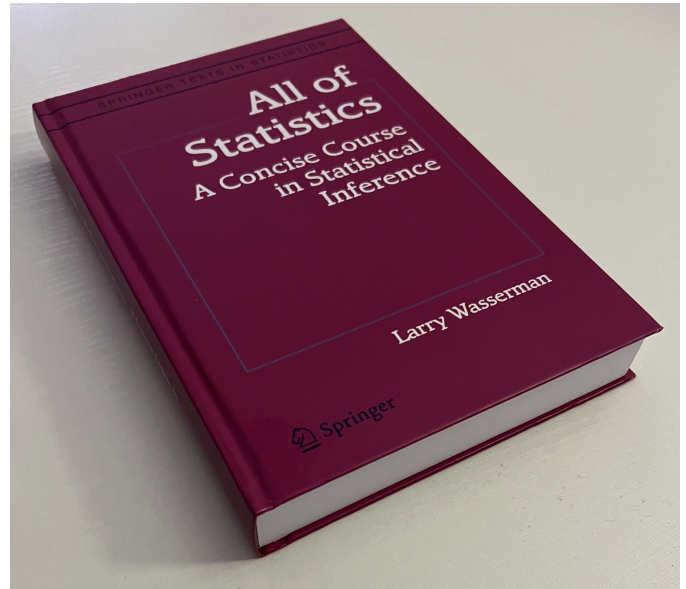
Code-only answers do not need to be duplicated in the PDF. We will grade those directly from the submitted notebook.

# Task1 PDF Contents

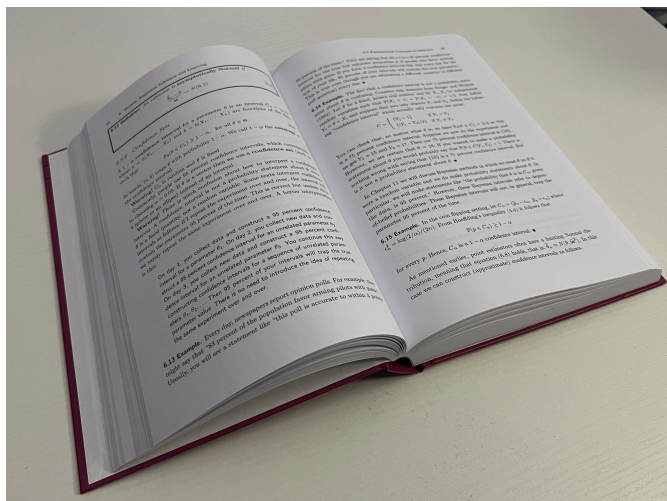
## 0.1 Three photographs



mild view: loaded 886 × 1200



strong view: loaded 1200 × 1054



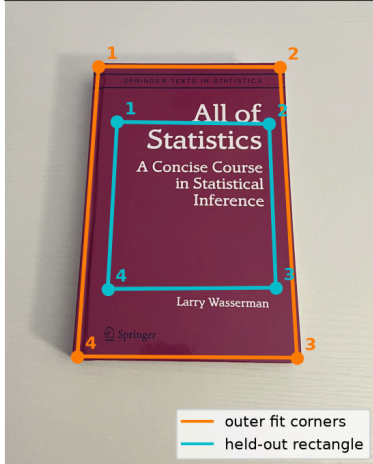
failure.jpg

# Task1 PDF Contents

## 0.2 Screenshots of the interactive affine and homography canvas

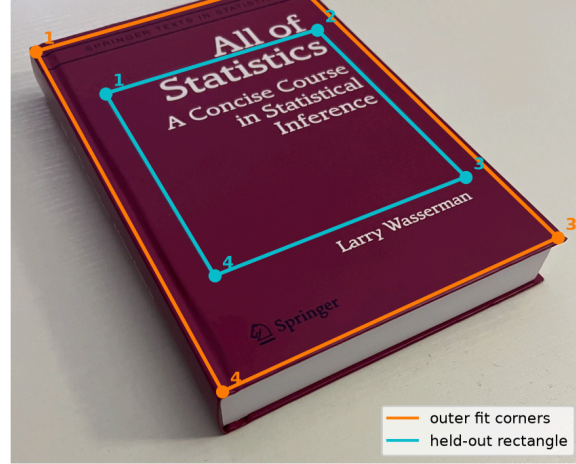
mild: source

inner: 4/4 points | top-left, then clockwise

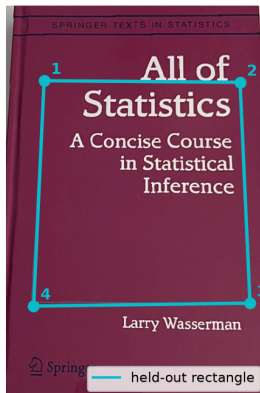


strong: source

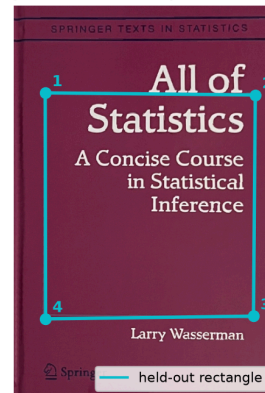
inner: 4/4 points | top-left, then clockwise



Affine least-squares fit  
right-angle error 2.62°; parallel error 3.25°

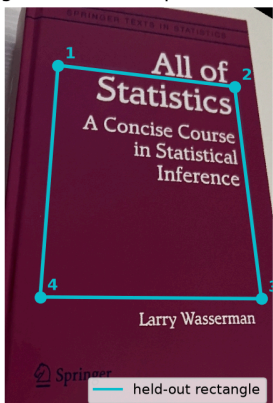


Homography fit  
right-angle error 0.65°; parallel error 0.93°

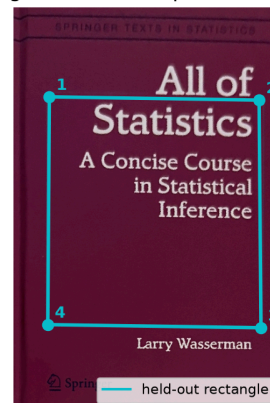


mild:

Affine least-squares fit  
right-angle error 7.05°; parallel error 9.01°



Homography fit  
right-angle error 0.91°; parallel error 0.45°



strong:

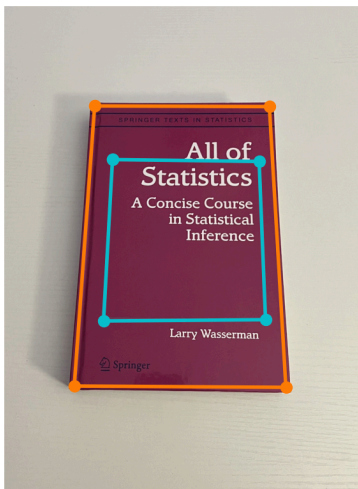
## Task1 PDF Contents

### 0.3 The rectification baseline comparison plot

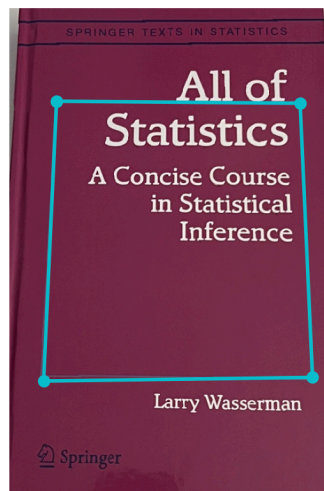
```
1 homework1_baseline_results = show_rectification_baseline(homework1_baseline_results  
2     homework1_comparison_state  
3 )  
✓ [7] 356毫秒
```

| view   | model      | right-angle error | parallel error |
|--------|------------|-------------------|----------------|
| mild   | affine     | 2.624             | 3.249          |
| mild   | homography | 0.651             | 0.935          |
| strong | affine     | 7.046             | 9.012          |
| strong | homography | 0.906             | 0.454          |

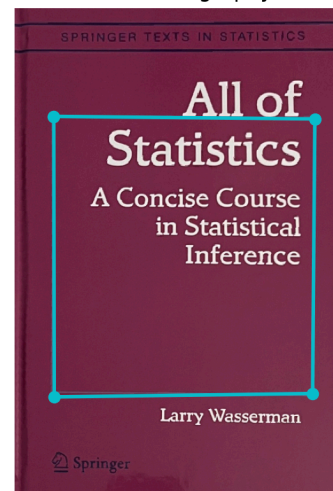
mild: selected source



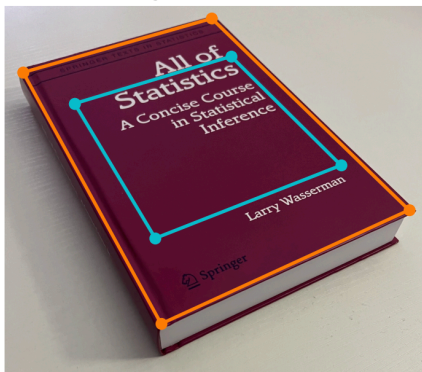
mild: affine



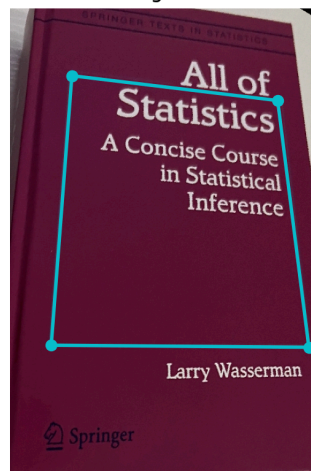
mild: homography



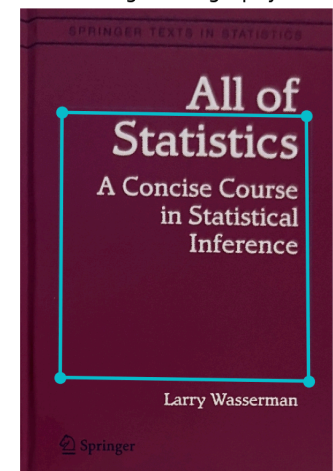
strong: selected source



strong: affine



strong: homography



# Task1 PDF Contents

## 0.4 The corner jitter diagnostic plot

## 0.5 The corner uncertainty results plot

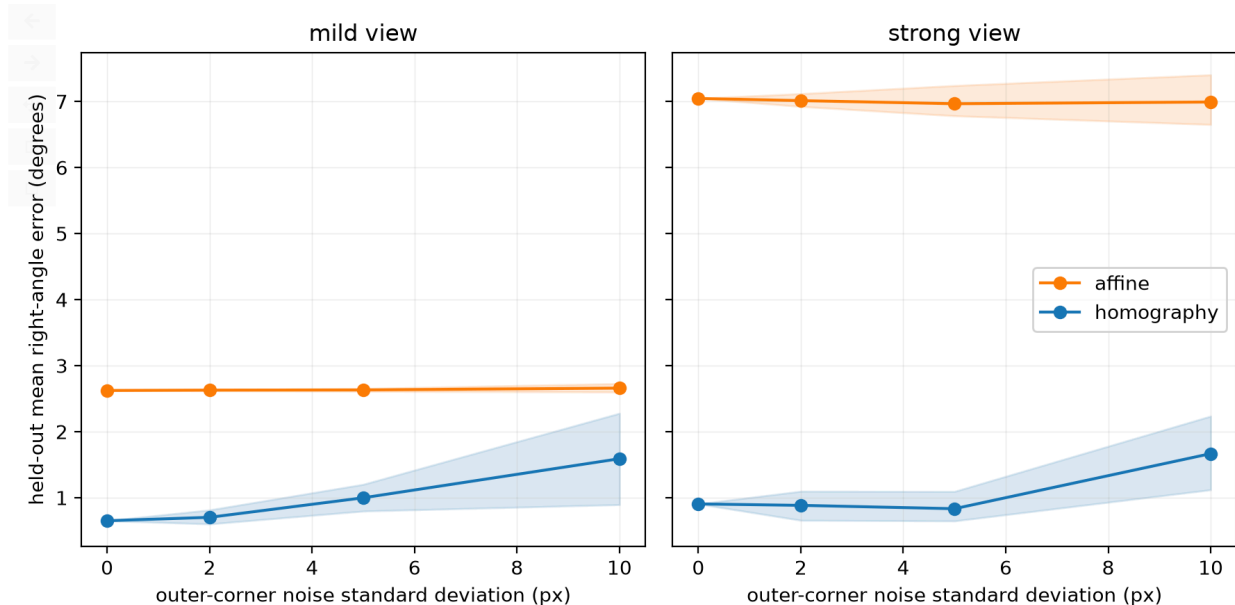
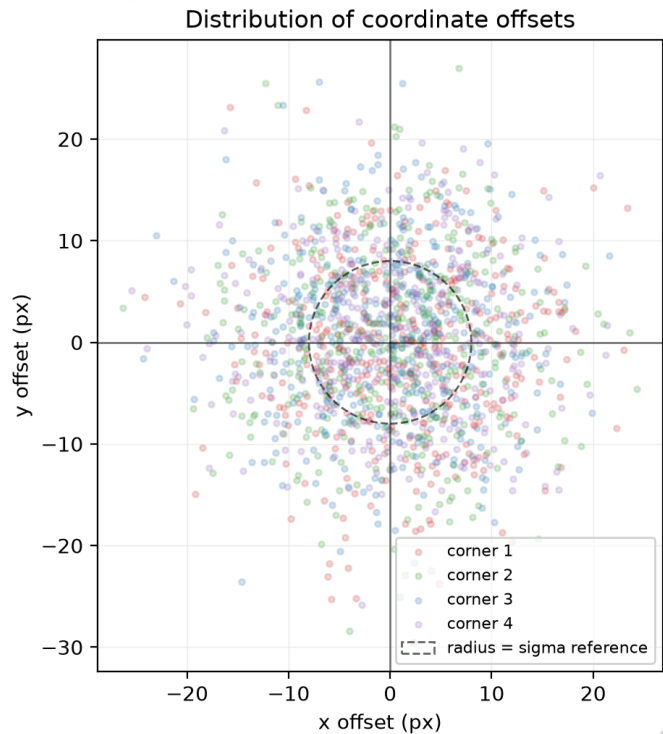
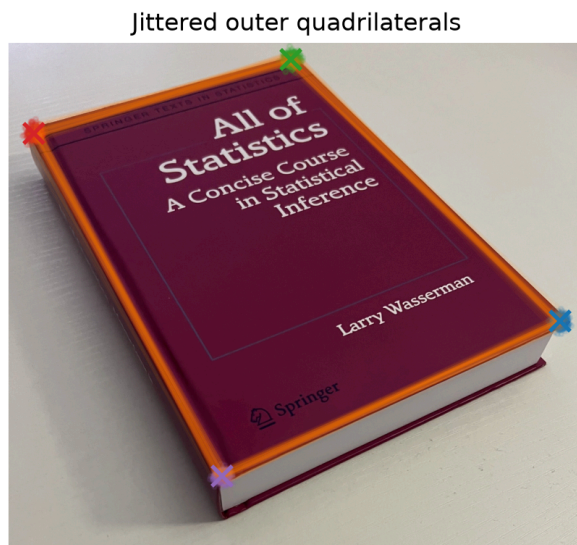
Requested sigma: 8.00 px

Empirical mean offset [dx, dy]: [0.281 0.064] px

Empirical standard deviation [dx, dy]: [8.027 8.213] px

Largest absolute cross-coordinate correlation: 0.087

Distribution check: plausible independent zero-mean Gaussian jitter.

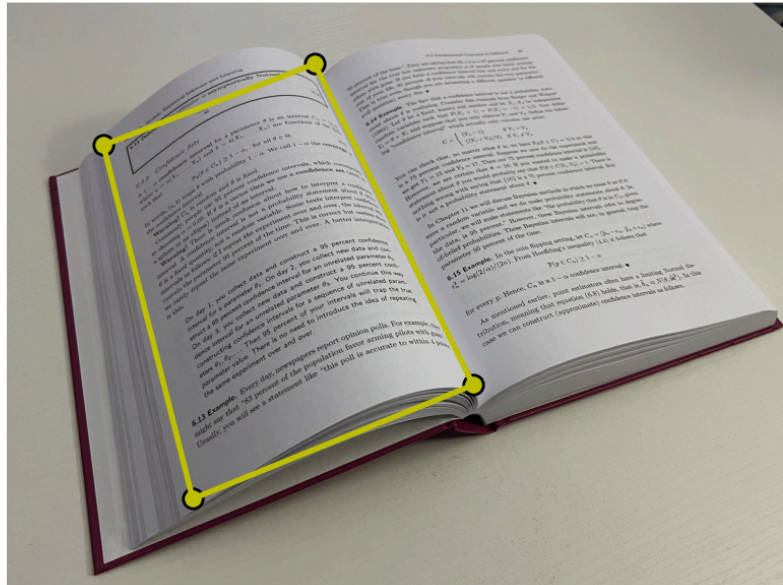


## Task1 PDF Contents

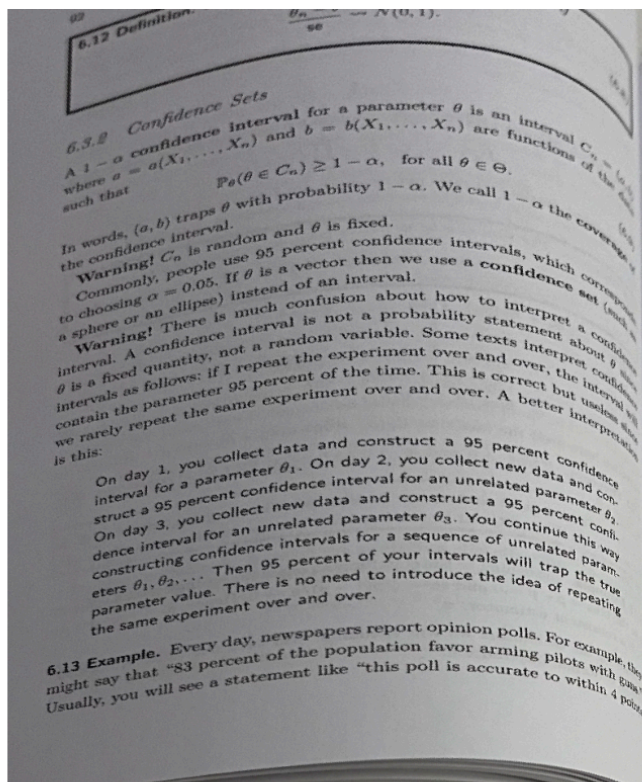
0.6 failure.jpg with its four selected corners visible

0.7 The rectified output for failure.jpg

Click 4 corners, then drag to refine



Rectified plane



# Task1 PDF Contents

## 0.8 Written answers for Tasks 1.2 1.3 and 1.4

### 1.2 Viewpoint comparison [0.75 marks]

For **each of the two planar photographs**, discuss the differences between affine and homography rectification visually and also error-wise. Then explain how moving from the mild to the strong viewpoint changed the difference between the models.

**1.2 Answer:** *In the mild view, the two rectifications look broadly similar, but homography restores the inner rectangle more accurately, with lower right-angle and parallel errors ( $0.651^\circ$  and  $0.935^\circ$ ) than affine rectification ( $2.624^\circ$  and  $3.249^\circ$ ). In the strong view, affine rectification leaves more noticeable perspective distortion, while homography produces a more rectangular result, reducing the respective errors from  $7.046^\circ$  and  $9.012^\circ$  to  $0.906^\circ$  and  $0.454^\circ$ . Thus, moving from the mild to the strong viewpoint increases the gap between the models because homography accounts for perspective distortion that an affine transformation cannot fully correct.*

### 1.3 Corner sensitivity [0.75 marks]

Discuss whether the same coordinate uncertainty had the same effect across models and viewpoints. Identify any result that did or did not match your initial expectation and give a plausible geometric or measurement-based explanation.

**1.3 Answer:** *The same corner noise affected the two models differently. Overall, homography's errors increased more with noise and showed greater variability, while affine medians remained fairly stable. This may be because homography fits the noisy corners exactly, whereas affine uses a least-squares fit with fewer degrees of freedom. Despite this greater sensitivity, homography retained lower median right-angle errors in both views. As expected, affine rectification showed greater variability in the strong view: at 10 px noise, its IQR was  $0.751^\circ$ , compared with  $0.127^\circ$  in the mild view. Stronger foreshortening can amplify corner-location errors during rectification, making the recovered angles more sensitive to the same pixel uncertainty.*

### 1.4 Homography failure case [0.75 marks]

Identify specific evidence in the rectified output that it failed. Explain which assumption of a single homography was violated and why changing only the corners cannot fix the complete result.

**1.4 Answer:** *In the rectified output, the definition-box borders and text lines near the spine remain curved. The page is physically curved, violating the assumption that the entire region lies on a single plane. Adjusting the corners changes the global homography but cannot remove the varying distortion within the page, so it cannot straighten all borders and text lines simultaneously.*

## Task2 PDF Contents

### 0.9 A screenshot of the interactive canvas

Active: Horizontal lines: right facade

Horizontal lines: left facade: 4 line(s);  $VP=(-16727.2, 3647.1)$   
Horizontal lines: right facade: 4 line(s);  $VP=(5055.9, 2915.3)$   
Red line: horizon from the two facade vanishing points  
Next click: endpoint 1



## Task2 PDF Contents

### 0.10 Derivations and answers for Tasks 2.2, 2.3, 2.4, and 2.8

#### 2.2 Inverse intrinsic matrix and rays [0.20 marks]

1. Write  $K^{-1}$ .
2. Compute the two ray vectors in terms of  $u_A, y_A, u_B, y_B, c_x, c_y$ , and  $f_{pixel}$ .

Your  $K^{-1}$  and ray expressions:

$$K^{-1} = \begin{bmatrix} 1/f_{pixel} & 0 & -c_x/f_{pixel} \\ 0 & 1/f_{pixel} & -c_y/f_{pixel} \\ 0 & 0 & 1 \end{bmatrix}.$$

Choosing  $\mathbf{r}_i = K^{-1}\tilde{\mathbf{v}}_i$ , the ray vectors are

$$\mathbf{r}_A = \begin{bmatrix} (u_A - c_x)/f_{pixel} \\ (y_A - c_y)/f_{pixel} \\ 1 \end{bmatrix}, \quad \mathbf{r}_B = \begin{bmatrix} (u_B - c_x)/f_{pixel} \\ (y_B - c_y)/f_{pixel} \\ 1 \end{bmatrix}.$$

#### 2.3 Ray constraint [0.30 marks]

3. Identify geometric constraint on the rays.
4. Expand the constraint and rearrange it to isolate  $f^2$ .

Your expressions:

The two facade directions are perpendicular in 3D, so  $\mathbf{r}_A^T \mathbf{r}_B = 0$ . Expanding gives

$$\frac{(u_A - c_x)(u_B - c_x) + (y_A - c_y)(y_B - c_y)}{f_{pixel}^2} + 1 = 0.$$

Multiplying by  $f_{pixel}^2$  and rearranging gives the following expressions.

•  $f_{pixel}^2$ :

$$f_{pixel}^2 = -[(u_A - c_x)(u_B - c_x) + (y_A - c_y)(y_B - c_y)].$$

•  $f_{pixel}$ :

$$f_{pixel} = \sqrt{-[(u_A - c_x)(u_B - c_x) + (y_A - c_y)(y_B - c_y)]}.$$

#### 2.4 Physical-validity condition [0.15 marks]

State the condition under which the recovered focal length is real. What would a negative value of  $f^2$  tell you about the selected lines or model assumptions?

- Condition under which the recovered focal length is real: The square-root argument must be non-negative. A physically valid, positive focal length requires  $f_{pixel}^2 > 0$ , equivalently

$$(u_A - c_x)(u_B - c_x) + (y_A - c_y)(y_B - c_y) < 0.$$

- Meaning of a negative value of  $f^2$ : No real focal length satisfies the orthogonality constraint under the assumed camera model. This may result from inaccurate endpoint selection, incorrectly grouped lines, non-perpendicular world directions, or invalid camera assumptions.

## Task2 PDF Contents

### 0.11 Derivations and answers for Tasks 2.2, 2.3, 2.4, and 2.8

### 0.12 Final focal-length results in pixels and millimetres

#### 2.8 Pixel-to-millimetre conversion [0.20 marks]

The focal lengths  $f_{px}$  and  $f_{mm}$  represent the **same distance on the camera's image plane**, expressed in different units.

For this uncropped portrait image, the full image width  $W_{px}$  pixels spans the sensor width  $S_x = 23.8$  mm. Let  $p_x$  denote the physical width represented by one horizontal pixel, measured in mm/pixel. The full sensor width therefore satisfies

$$(W_{px} \text{ pixels})(p_x \text{ mm/pixel}) = S_x \text{ mm}.$$

The focal length  $f_{px}$  spans  $f_{px}$  pixels along the same image-plane axis, so

$$(f_{px} \text{ pixels})(p_x \text{ mm/pixel}) = f_{mm} \text{ mm}.$$

Find  $p_x$  and  $f_{mm}$ . Implement using `taito_image.shape[1]` for  $W_{px}$  and your estimated `taito_focal_length_px` for  $f_{px}$ .

Your expression for  $p_x$ :

$$p_x = \frac{S_x}{W_{px}} = \frac{23.8}{W_{px}} \text{ mm/pixel}.$$

Your expression for  $f_{mm}$ :

$$f_{mm} = f_{px} p_x = f_{px} \frac{23.8}{W_{px}}.$$

Estimated focal length: 8136.9 px

Image scale: 0.009246 mm/pixel

Estimated focal length: 75.2 mm